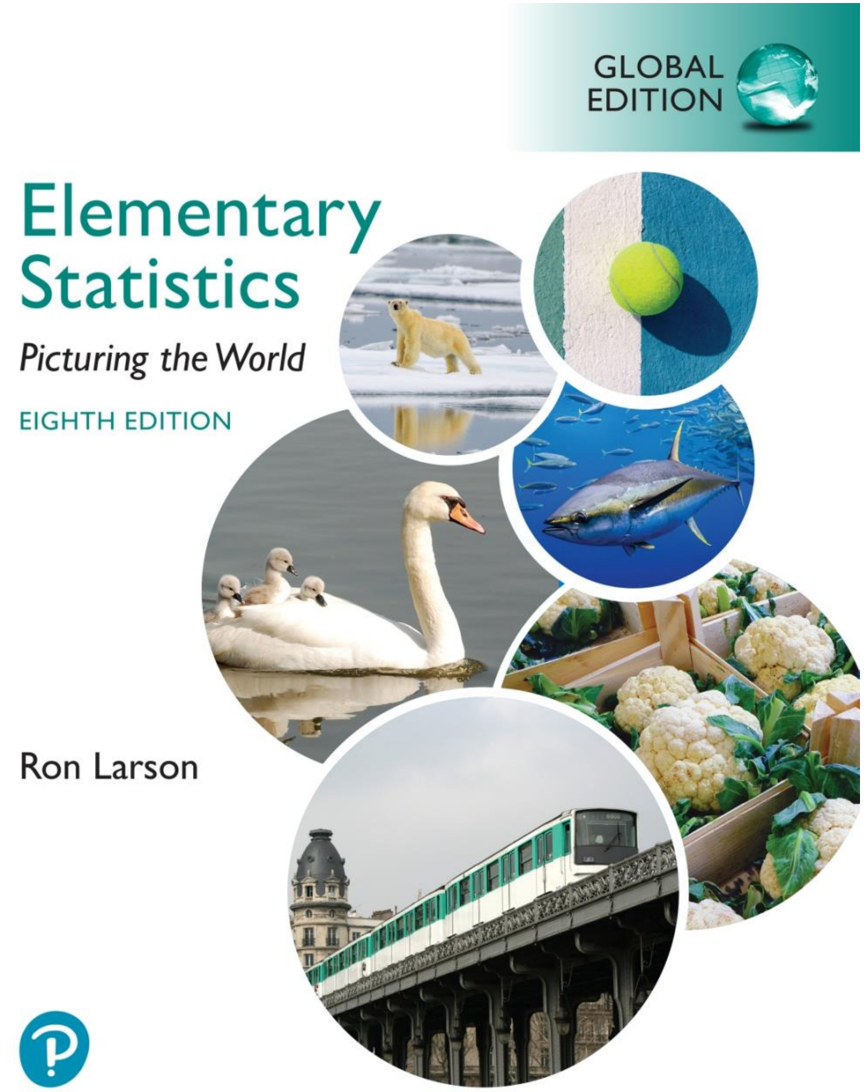
Chapter 3

Probability



Chapter Outline

- 3.1 Basic Concepts of Probability and Counting
- 3.2 Conditional Probability and the Multiplication Rule
- 3.3 The Addition Rule
- 3.4 Additional Topics in Probability and Counting

Section 3.3

The Addition Rule

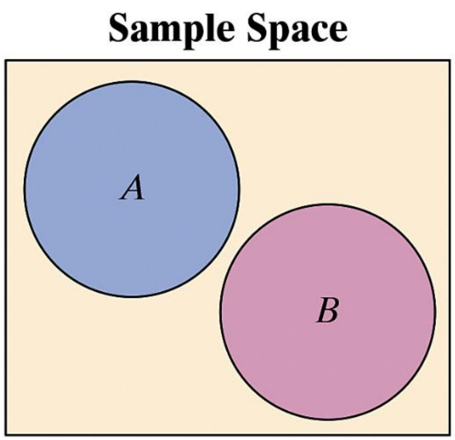
Section 3.3 Objectives

- How to determine whether two events are mutually exclusive
- How to use the Addition Rule to find the probability of two events

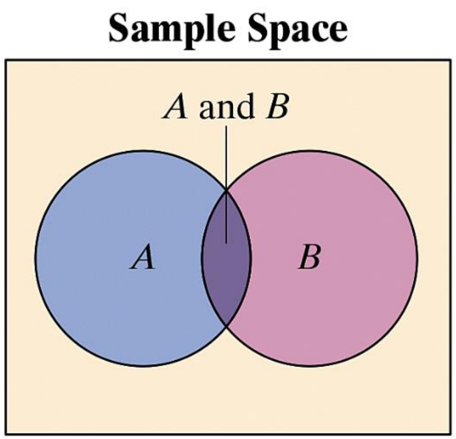
Mutually Exclusive Events

Mutually exclusive

- Two events *A* and *B* cannot occur at the same time
- *A* and *B* have no outcomes in common



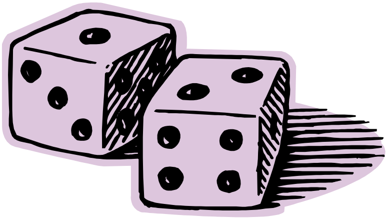
A and *B* are mutually exclusive



A and *B* are not mutually exclusive

Example: Recognizing Mutually Exclusive Events

Determine whether the events are mutually exclusive. Explain your reasoning.



1. Event A: Roll a 3 on a die.
   Event B: Roll a 4 on a die.

Solution:

Mutually exclusive (The first event has one outcome, a 3. The second event also has one outcome, a 4. These outcomes cannot occur at the same time.)

Example: Recognizing Mutually Exclusive Events

Determine whether the events are mutually exclusive. Explain your reasoning.

2. Event A: Randomly select a male student.
   Event B: Randomly select a nursing major.

Solution:

Not mutually exclusive (The student can be a male nursing major.)

Example: Recognizing Mutually Exclusive Events

Determine whether the events are mutually exclusive. Explain your reasoning.

3. Event *A*: Randomly select a blood donor with type O blood.
   Event *B*: Randomly select a female blood donor..

Solution:

Not mutually exclusive (The donor can be a female with type O blood)

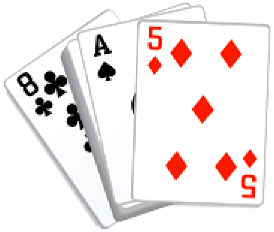
The Addition Rule

Addition rule for the probability of *A* or *B*

- The probability that events *A* or *B* will occur is
  - $\mathbf{P}(A \textbf{ or } B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \textbf{ and } B)$
- For mutually exclusive events *A* and *B*, the rule can be simplified to
  - $\mathbf{P}(A \textbf{ or } B) = \mathbf{P}(A) + \mathbf{P}(B)$
  - Can be extended to any number of mutually exclusive events

Example: Using the Addition Rule to Find Probabilities

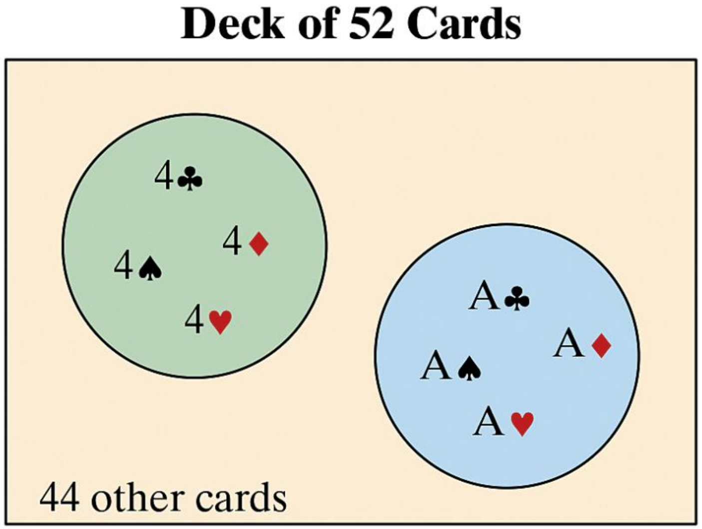
1. You select a card from a standard deck. Find the probability that the card is a 4 or an ace.



Solution:

The events are mutually exclusive (if the card is a 4, it cannot be an ace)

$$P(4\ or\ ace) = P(4) + P(ace)$$
$$= \frac{4}{52} + \frac{4}{52}$$
$$= \frac{8}{52} \approx 0.154$$

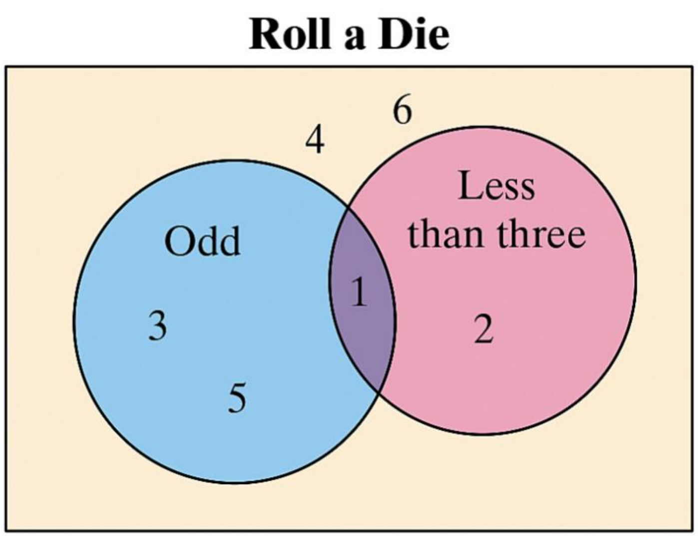


Example: Using the Addition Rule to Find Probabilities

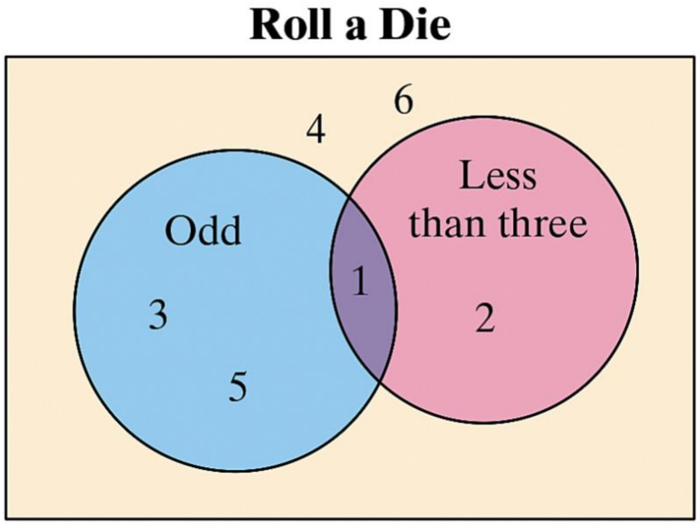
2. You roll a die. Find the probability of rolling a number less than 3 or rolling an odd number.

Solution:

The events are not mutually exclusive (1 is an outcome of both events)



Solution: Using the Addition Rule to Find Probabilities



$$P(\text{less than } 3 \text{ or odd})$$

$$= P(\text{less than } 3) + P(\text{odd}) - P(\text{less than } 3 \text{ and odd})$$

$$= \frac{2}{6} + \frac{3}{6} - \frac{1}{6} = \frac{4}{6} \approx 0.667$$

Example: Finding Probabilities of Mutually Exclusive Events

The frequency distribution shows volumes of sales (in dollars) and the number of months in which a sales representative reached each sales level during the past three years. Using this sales pattern, find the probability that the sales representative will sell between $75,000 and $124,999 next month.

Sales volume ($)	Months
0–24,999	3
25,000–49,999	5
50,000–74,999	6
75,000–99,999	7
100,000–124,999	9
125,000–149,999	2
150,000–174,999	3
175,000–199,999	1

Solution: Finding Probabilities of Mutually Exclusive Events

- A = {monthly sales between \$75,000 and \$99,999}
- B = {monthly sales between \$100,000 and \$124,999}
- A and B are mutually exclusive

Sales volume (\$)	Months
0–24,999	3
25,000–49,999	5
50,000–74,999	6
75,000–99,999	7
100,000–124,999	9
125,000–149,999	2
150,000–174,999	3
175,000–199,999	1

Solution: Finding Probabilities of Mutually Exclusive Events

- A = {monthly sales between \$75,000 and \$99,999}
- B = {monthly sales between \$100,000 and \$124,999}
- A and B are mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B)$$

$$= \frac{7}{36} + \frac{9}{36}$$

$$= \frac{16}{36} \approx 0.444$$

Sales volume (\$)	Months
0–24,999	3
25,000–49,999	5
50,000–74,999	6
75,000–99,999	7
100,000–124,999	9
125,000–149,999	2
150,000–174,999	3
175,000–199,999	1

Example: Using the Addition Rule to Find Probabilities

A blood bank catalogs the types of blood, including whether it is Rh-positive or Rh-negative, given by donors during the last five days. The number of donors who gave each blood type is shown in the table.

1. Find the probability the donor has type O or type A blood.

	Type O	Type A	Type B	Type AB	Total
Rh-Positive	156	139	37	12	344
Rh-Negative	28	25	8	4	65
Total	184	164	45	16	409

Solution: Using the Addition Rule to Find Probabilities

The events are mutually exclusive (a donor cannot have type O blood and type A blood)

	Type O	Type A	Type B	Type AB	Total
Rh-Positive	156	139	37	12	344
Rh-Negative	28	25	8	4	65
Total	184	164	45	16	409

$$P(\textit{type O or type A}) = P(\textit{type O}) + P(\textit{type A})$$

$$= \frac{184}{409} + \frac{164}{409}$$

$$= \frac{348}{409} \approx 0.851$$

Example: Using the Addition Rule to Find Probabilities

2. Find the probability the donor has type B or is Rh-negative.

	Type O	Type A	Type B	Type AB	Total
Rh-Positive	156	139	37	12	344
Rh-Negative	28	25	8	4	65
Total	184	164	45	16	409

Solution:

The events are not mutually exclusive (a donor can have type B blood and be Rh-negative)

Solution: Using the Addition Rule to Find Probabilities

	Type O	Type A	Type B	Type AB	Total
Rh-Positive	156	139	37	12	344
Rh-Negative	28	25	8	4	65
Total	184	164	45	16	409

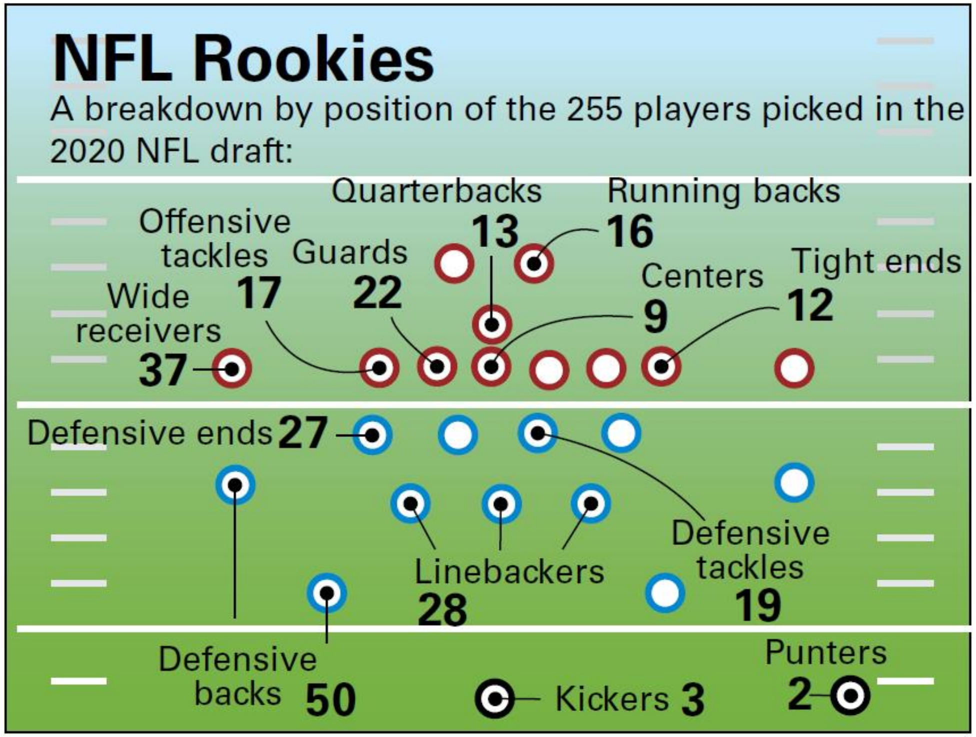
$$P(type\ B\ or\ Rh-neg)$$
$$= P(type\ B) + P(Rh-neg) - P(type\ B\ and\ Rh-neg)$$
$$= \frac{45}{409} + \frac{65}{409} - \frac{8}{409} = \frac{102}{409} \approx 0.249$$

A Summary of Probability

Type of probability and probability rules	In words	In symbols
Classical Probability	The number of outcomes in the sample space is known and each outcome is equally likely to occur.	$P(E) = \dfrac{\text{Number of outcomes in event } E}{\text{Number of outcomes in sample space}}$
Empirical Probability	The frequency of each outcome in the sample space is estimated from experimentation.	$P(E) = \dfrac{\text{Frequency of event } E}{\text{Total frequency}} = \dfrac{f}{n}$
Range of Probabilities Rule	The probability of an event is between 0 and 1, inclusive.	$0 \le P(E) \le 1$
Complementary Events	The complement of event E is the set of all outcomes in a sample space that are not included in E, and is denoted by E'.	$P(E') = 1 - P(E)$
Multiplication Rule	The Multiplication Rule is used to find the probability of two events occurring in sequence.	$P(A \text{ and } B) = P(A) \cdot P(B \mid A)$ Dependent events $P(A \text{ and } B) = P(A) \cdot P(B)$ Independent events
Addition Rule	The Addition Rule is used to find the probability of at least one of two events occurring.	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ $P(A \text{ or } B) = P(A) + P(B)$ Mutually exclusive events

Example: Combining Rules to Find Probabilities

Use the figure to find the probability that a randomly selected draft pick is not a running back or a wide receiver.



Solution: Combining Rules to Find Probabilities

Solution:

Define events A and B.

A: Draft pick is a running back.

B: Draft pick is a wide receiver.

These events are mutually exclusive, so the probability that the draft pick is a running back or wide receiver is

$$P(A \text{ or } B) = P(A) + P(B) = \frac{16}{255} + \frac{37}{255} = \frac{53}{255}.$$

Solution: Combining Rules to Find Probabilities

Solution:

By taking the complement of $P(A \text{ or } B)$, you can determine that the probability of randomly selecting a draft pick who is not a running back or wide receiver is

$$1 - P(A \text{ or } B) = 1 - \frac{53}{255} = \frac{202}{255} \approx 0.792.$$